

Boundary Value Analysis

Boundary Value Analysis is a software testing technique in which test cases are designed around the minimum and maximum limits of an input field. Values at, just below, and just above the boundaries are tested to identify defects.

Fields tested:

1. Age: 18–60
2. Password: 8–16 characters
3. Transaction Amount: ₹100–₹50,000

Total test cases: 18

1. Age Field

Requirement: Age must be between **18 and 60**.

Test Case	Input	Expected Result
BVA-01	17	Invalid
BVA-02	18	Valid
BVA-03	19	Valid
BVA-04	59	Valid
BVA-05	60	Valid
BVA-06	61	Invalid

Boundary values: 18 and 60.

2. Password Length

Requirement: Password must contain **8–16 characters**.

Test Case	Input length	Expected Result
BVA-07	7	Invalid
BVA-08	8	Valid
BVA-09	9	Valid

Test Case	Input length	Expected Result
BVA-10	15	Valid
BVA-11	16	Valid
BVA-12	17	Invalid

Boundary values: 8 and 16 characters.

3. Transaction Amount

For our mobile banking app, let's assume:

Requirement: Transaction amount must be between **₹100 and ₹50,000**.

Test Case	Amount	Expected Result
BVA-13	₹99	Invalid
BVA-14	₹100	Valid
BVA-15	₹101	Valid
BVA-16	₹49,999	Valid
BVA-17	₹50,000	Valid
BVA-18	₹50,001	Invalid

Boundary values: ₹100 and ₹50,000.